LESSON 7.4

Compare and Order Integers and Rational Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can compare and order integers using a number line.

WHY THIS MATTERS Lucy and Yuzuki are playing a trivia game with these rules: if a player answers a question correctly, they receive 1 point; if they answer incorrectly, they lose 0.25 point. The player with the greatest score wins the game. The final scores for two games are shown in the table. Who is the winner of each game?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order; numbers get smaller to the left and larger to the right.	left is less, right is more
Compare *Spanish: Comparar*	To see if a number is bigger, smaller, or equal to another.	−7 compared with −3
Order *Spanish: Ordenar*	To arrange numbers by size, from least to greatest or greatest to least.	Least to greatest: -5, -2, 0, 3, 7 (left to right on the number line)
Greater than *Spanish: Mayor que*	The > sign, showing one number is bigger.	−3 > −7
Less than *Spanish: Menor que*	The < sign, showing one number is smaller.	−7 < −3
Negative integer *Spanish: Entero negativo*	A whole number smaller than zero, like -5.	−7, −3, −100

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Compare −14 degrees and −5 degrees.

1. Plot −14 and −5 on a number line.
2. −5 is to the right of −14.
3. Write the comparison: −14 < −5.

Answer: −5 is greater; −14 is colder

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's compare -2 and 3. Which one is to the left on the number line? The -2 is left of zero.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- On the number line, _____ sits to the _____ of _____ , so _____ is greater.
- I can write the comparison as _____ _____ _____ .
- The least number in this set is _____ because _____ .
- Comparing two negative numbers is different from counting because _____ .

Word bank compare | order | greater than | less than | number line | left | right | negative | positive | least | greatest | inequality

WATCH OUT!

A common mistake

A common mistake in Compare and Order Integers is thinking that a negative number with a bigger digit is the bigger value

LESSON 7.4

Compare and Order Integers and Rational Numbers

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement is TRUE?

(A) -5 < -2

(B) -4 > 0

(C) -3 > -1

(D) -1 < -6

HINT Sketch a quick number line. For negatives, 'greater' means closer to zero, not a bigger digit.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Drag to order these integers from LEAST to GREATEST.

Groups:

_____ 3

_____ -1

_____ -6

_____ 0

_____ -4

_____ 5

HINT Least to greatest reads left to right on a number line — all the negatives come before 0.

3 **Mark and label each value on the number line.**

Plot -3, 1, -7, 4, and -1 on the number line to help you compare them.



HINT Negatives (-3, -7, -1) go left of 0; positives (1, 4) go right.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Which integer is the LEAST: -3, 5, -8, 2, 0?

(A) -8
(B) -3
(C) 0
(D) 5

HINT 'Least' means farthest LEFT on the number line — for negatives, that's the one farthest below zero.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is |−8|?

(A) 0
(B) 8
(C) −8
(D) −16

HINT The bars |...| mean absolute value: the distance from 0, which is never negative.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** Compare -5 and -2. Write the comparison using > or <.
2. **Attempt** The student wrote -5 > -2, reasoning that 5 is bigger than 2, so -5 must be bigger than -2.
3. **Check** On a number line, -5 is farther LEFT than -2, so -5 is actually less. The correct comparison is -5 < -2.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Where do -5 and -2 sit on the number line?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does plotting on the number line help you compare and order integers?

SENTENCE STARTER On the number line, ___ is to the left of ___, so ___ < ___. The order from least to greatest is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.4

Compare and Order Integers and Rational Numbers

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Plot these integers and write them in order from greatest to least: -8, 4, -3, 1, -1, 6



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

Compare each pair of integers using < , > , or =.

Integer A	Integer B	Comparison (< , > , or =)
-4	2	
-7	-3	
0	-5	
-1	-1	
6	-9	

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

Sort these integers from GREATEST to LEAST.

Groups:

_____ -10

_____ 5

_____ -2

_____ 8

_____ 0

_____ -6

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Three cities have temperatures of -12°F, -5°F, and 3°F. Which city is warmest?

(A) -5°F

(B) -12°F

(C) 3°F

(D) They are equal

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read** Order -8 and -15 from least to greatest.
2. **Attempt** The student wrote -8, -15, because 8 is smaller than 15.
3. **Check** -15 is farther left than -8 on the number line, so -15 is the least. The correct order is -15, -8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does plotting on the number line help you compare and order integers?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 7.4

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Challenge · Extension

MASTERY CHECK
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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

These integers represent bank account changes. Sort from the WORST change (biggest loss) to the BEST change (biggest gain).

Groups:

_____ +50 deposit

_____ -25 withdrawal

_____ +10 deposit

_____ -100 withdrawal

_____ 0 no change

_____ -5 fee

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Comparison Error

1. **Problem** Compare -3 and -8. Which is greater?
2. **Student thinking** 8 is bigger than 3, so -8 must be greater than -3
3. **Student answer** -8 > -3

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A student says: 'Negative numbers are always less than positive numbers, so -100 is less than 1.' Is this correct? Now the student says: '-100 is less than -1 because 100 is greater than 1.' Is THIS correct? Explain the difference.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How does plotting on the number line help you compare and order integers?

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